

# NISHANTH VEDURUVADA

## Principal AI & Security Architect

Agentic AI • Enterprise AI Modernization • AI Governance • Zero Trust Architecture

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### PROFILE

Architect specialising in enterprise AI systems built under Zero Trust principles. Currently AI Engineer Advisor at NTT DATA, leading architecture and technical direction across 8–10 AI modernization programs in the Syntphony product ecosystem, on platforms serving approximately 2 million users, with a team of eight engineers and data scientists. Previously sole architect of the AI Trust Fabric for a national-scale insurance data ecosystem — work that became a registered design and a published paper. Depth spans agentic systems and tool protocols (MCP, A2A), enterprise retrieval, AI governance, and NIST 800-207 Zero Trust, with an unusual concentration in regulated insurance on both the security and platform-modernization sides.

### EXPERIENCE

#### NTT DATA

Dec 2025 – Present

##### AI Engineer Advisor

*Architecture and technical governance for AI modernization across NTT DATA's Syntphony ecosystem — Syntphony AI, Open Telecom Networks, Insurance Cloud Migration, Stations, and Commerce & Payments.*

- **Agentic Platform Architecture:** Defined the reference architecture for a reusable multi-agent platform, standardising on MCP for tool integration and A2A for cross-domain agent collaboration, so domain teams inherit governance rather than rebuilding it per engagement.
- **AI Governance Controls:** Established per-agent OAuth2 identity, full tool-call audit logging, configurable rate limits, action-scope enforcement, and a human-in-the-loop approval gateway for high-risk actions — inherited automatically by every domain agent on the platform.
- **Enterprise Retrieval Architecture:** Designed a dual-path knowledge assistant separating live analytics from indexed content: semantic-model queries with row-level security passed through as the calling user, alongside permission-scoped document retrieval, routed by a query classifier. Metric values are never embedded, so answers cannot go stale.
- **Legacy Modernization:** Architected a RAG system over mainframe COBOL and JCL estates for insurance cloud migration, built on a structure-aware parser and static dependency graph. Chose retrieval over fine-tuning so every answer cites program, paragraph and line.
- **Cross-Domain Operational Intelligence:** Designed multi-agent orchestration across five OSS domains — RAN, Core, Transport, Inventory and Assurance — using domain-specialist agents under a query planner, with a correlation layer producing evidence-cited answers rather than model summaries.
- **Conversational Analytics:** Designed an embedded analytics assistant over existing semantic models, with multi-tenant isolation via Entra ID and row-level security scoped per business unit and country, avoiding duplication of the client's existing BI investment.
- **Technical Leadership:** Lead architecture reviews, decision records and trade-off analysis for a team of eight; own enterprise AI architecture standards, technical strategy, stakeholder alignment and mentoring.

**AI Enterprise Architect**

Client: Insurance Information Bureau of India (IIB)

*Sole architect of the AI Trust Fabric for a national-scale insurance data ecosystem aggregating records from every insurance company in India, under IRDAI regulation and data-localisation requirements.*

- **Zero Trust Architecture:** Designed the full NIST 800-207 structure — policy administration, decision, enforcement and information points as separate concerns — with enforcement deployed as an API gateway intercept so every call, human or agent, traverses it.
- **Adaptive Trust Engine:** Replaced binary allow/deny with continuous trust scoring across identity confidence, device posture, behavioural baseline, data classification and context, driving allow / step-up-authentication / deny decisions with thresholds configurable per data class.
- **AI Agent Identity & Governance:** Gave agents their own workload identity rather than borrowed user credentials, with action-scope enforcement, per-agent rate limiting, prompt-injection detection, full decision audit, and human escalation above a risk threshold.
- **Offline AI Capability:** Architected fully offline inference with Ollama to satisfy no-internet-egress environments — a hard constraint that would otherwise have blocked the entire governance layer.
- **RAG Policy Intelligence:** Built natural-language access to IRDAI regulations and internal policy, orchestrated with LangGraph and CrewAI, with grounded and cited responses for compliance analysts.
- **Explainability:** Wired traceable rationale into every access decision — which policy matched, which trust signals fired — making AI-assisted decisions defensible under regulatory audit.

**Arcstream Technologies**

Aug 2022 – Jan 2025

**Technical Security Lead**

- **Enterprise Security Modernization:** Led Zero Trust network segmentation and a SASE migration covering 5,000+ users, replacing perimeter-based access with identity-centric controls.
- **Secure Delivery:** Hardened CI/CD pipelines and established architecture governance across engineering teams.

**Reezanorp Technologies**

Dec 2016 – Jul 2022

**Senior Software Engineer**

- **Applied ML & Data Engineering:** Built machine learning, analytics, computer vision, ETL and enterprise automation systems in Python — the engineering foundation underneath the later architecture work.

**PUBLICATIONS & INTELLECTUAL PROPERTY**

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**MCP Guardian: A Security-First Layer for Safeguarding MCP-Based AI Systems**

Computer Science & Information Technology (CS&IT), Vol.15 No.9, pp.107–121, 2025 · DOI 10.5121/csit.2025.150908

Middleware for the Model Context Protocol adding authentication, per-token rate limiting, WAF-style scanning of tool calls and request logging, without modifying individual MCP servers. Evaluated at roughly 3–4 ms median latency overhead.

**Building the Trust Fabric: Reinventing National-Scale Insurance Security in the AI Era**

Research in progress

**Secure Insurance Data Management Device**

Registered Design No. 485611-001, Government of India · Co-inventor

Secure insurance data management architecture incorporating Zero Trust principles, adaptive trust evaluation, identity-centric controls and secure data access mechanisms.

**ARCHITECTURE COMPETENCIES**

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**Identity & Authorization** — agent workload identity, OAuth2/OIDC, Entra ID, ZTNA, IAM, RLS passthrough

**Agentic Systems** — MCP, FastMCP, A2A, CrewAI, LangGraph, AutoGen, Semantic Kernel, human-in-the-loop

**Governance & Observability** — action-scope enforcement, audit trails, rate limiting, OpenTelemetry, evaluation

**Policy & Trust** — NIST 800-207, PAP/PDP/PEP/PIP, adaptive trust scoring, step-up authentication

**Retrieval & Grounding** — RAG, hybrid and semantic search, vector stores, domain-aware parsing, citation design

**Platform & Delivery** — Azure, Azure OpenAI, AI Foundry, Kubernetes, event-driven architecture, Python, FastAPI, CI/CD

**EDUCATION**

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**B.Tech, Electronics & Communication Engineering** — Jawaharlal Nehru Technological University